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Overhead stowage bin with electronic control system

Abstract

A locking system for an overhead stowage bin may comprise a latch and an electromechanical actuator configured to actuate the latch. A bin controller may be electrically coupled to the electromechanical actuator. A passenger input and a master controller may be in operable communication with the bin controller.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of, claims priority to and the benefit of, U.S. patent application Ser. No. 17/136,381, filed Dec. 29, 2020, entitled “OVERHEAD STOWAGE BIN WITH ELECTRONIC CONTROL SYSTEM” (hereinafter the '381 Application). The '381 application claims priority to and the benefit of India patent Application No. 202041009538, filed Mar. 5, 2020 and titled “OVERHEAD STOWAGE BIN WITH ELECTRONIC CONTROL SYSTEM” (hereinafter the '538 Application). The '381 and '538 Applications are hereby incorporated by reference in their entirety for all purposes.

FIELD

(1) The present disclosure relates to stowage compartments, and, more specifically, to overhead stowage bins for aircraft.

BACKGROUND

(2) Aircraft generally include overhead stowage bins for passengers to stow their carry-on luggage during flight. Stowing carry-on items in the overhead stowage bin is limited by bin space and the size of carry-on luggage. Carry-on luggage is generally stored on a first-come first-serve basis. Boarding delays can occur when passengers boarding in later boarding groups have difficulty determining which bins are full and which bins have sufficient storage room remaining. Further, existing bins generally do not have a locking feature. During an emergency evacuation, people may try to access the bin and remove luggage, which can delay the evacuation procedure.

SUMMARY

(3) A locking system for an overhead stowage bin is disclosed herein. In accordance with various embodiments, the locking system may comprise a latch and an electromechanical actuator

configured to actuate the latch. A bin controller may be electrically coupled to the electromechanical actuator. A passenger input may be in operable communication with the bin controller. A master controller may be in operable communication to the bin controller.

(4) In various embodiments, a sensor may be in operable communication with the bin controller. The sensor may be configured to determine an amount of available space in the overhead stowage bin. In various embodiments, a display may be in operable communication with the bin controller.

(5) In various embodiments, a first tangible, non-transitory memory may be configured to communicate with the bin controller. The first tangible, non-transitory memory may have instructions stored thereon that, in response to execution by the bin controller, cause the bin controller to perform a first set of operations, which may comprise receiving, by the bin controller, a signal from the passenger input; determining, by the bin controller, if the locking system is in a locked state or an unlocked state; and determining, by the bin controller, whether to output an electrical signal to the electromechanical actuator based on whether the locking system is in the locked state or the unlocked state.

(6) In various embodiments, the first set of operations may further comprise receiving, by the bin controller, a lock command from the master controller; and determining, by the bin controller, not to output the electrical signal to the electromechanical actuator based on the lock command from the master controller.

(7) In various embodiments, the first set of operations may further comprise outputting, by the bin controller, the electrical signal to the electromechanical actuator if the bin controller determines the locking system is in the unlocked state.

(8) In various embodiments, a flight crew input may be in operable communication with the master controller. A second tangible, non-transitory memory may be configured to communicate with the master controller. The second tangible, non-transitory memory may have instructions stored thereon that, in response to execution by the master controller, cause the master controller to perform a second set of operations, which may comprise receiving, by the master controller, a lock signal from the flight crew input; and sending, by the master controller, a lock command to the bin controller.

(9) In various embodiments, a display may be in operable communication with the bin controller. The first set of operations may further comprise receiving, by the bin controller, the lock command from the master controller; and commanding, by the bin controller, the display to output a locked message.

(10) An overhead stowage bin is also disclosed herein. In accordance with various embodiments, the overhead stowage bin may, comprise a door configured for rotation about a hinge joint and a latch configured to secure the door in a closed position. An electromechanical actuator may be configured to actuate the latch. A bin controller may be electrically coupled to the electromechanical actuator. A master controller may be in operable communication with the bin controller.

(11) In various embodiments, a first tangible, non-transitory memory may be configured to communicate with the bin controller. The first tangible, non-transitory memory may have instructions stored thereon that, in response to execution by the bin controller, cause the bin controller to perform a first set of operations, which may comprise receiving, by the bin controller, a signal from a passenger input; determining, by the bin controller, if the overhead stowage bin is in a locked state or an unlocked state; and determining, by the bin controller, whether to output an electrical signal to the electromechanical actuator based on whether the overhead stowage bin is in the locked state or the unlocked state.

(12) In various embodiments, a second tangible, non-transitory memory may be configured to communicate with the master controller. The second tangible, non-transitory memory may have instructions stored thereon that, in response to execution by the master controller, cause the master controller to perform a second set of operations, which may comprise receiving, by the master

controller, a lock signal from a flight crew input; and sending, by the master controller, a lock command to the bin controller.

(13) In various embodiments, the first set of operations may further comprise determining, by the bin controller, not to output the electrical signal to the electromechanical actuator based on the lock command from the master controller.

(14) In various embodiments, a sensor may be in operable communication with the bin controller. The sensor may be configured to detect a condition within the overhead stowage bin.

(15) In various embodiments, a display may be located on the door. In various embodiments, the display may be configured to output at least one of a locked status or an available space status.

(16) An overhead stowage bin in accordance with various embodiments, may comprise a door configured for rotation about a hinge joint and an electromechanical actuator configured to rotate the door about the hinge joint. A bin controller may be electrically coupled to the electromechanical actuator. A master controller may be in operable communication with the bin controller.

(17) In various embodiments, a tangible, non-transitory memory may be configured to communicate with the bin controller. The tangible, non-transitory memory may have instructions stored thereon that, in response to execution by the bin controller, cause the bin controller to perform operations, which may comprise receiving, by the bin controller, a signal from a passenger input; determining, by the bin controller, if the overhead stowage bin is in a locked state or an unlocked state; and determining, by the bin controller, whether to output an electrical signal to the electromechanical actuator based on whether the overhead stowage bin is in the locked state or the unlocked state.

(18) In various embodiments, the electromechanical actuator may be configured to rotate the door about the hinge joint in response to receiving the electrical signal from the bin controller. In various embodiments, a sensor may be in operable communication with the bin controller. The sensor may be configured to detect an obstruction in a path of the door.

(19) In various embodiments, the instructions may further comprise receiving, by the bin controller, an output from the sensor; and determining, by the bin controller, whether to output the electrical signal to the electromechanical actuator based on the output from the sensor.

(20) The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be understood, however, the following description and drawings are intended to be exemplary in nature and non-limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the detailed description and claims when considered in connection with the figures, wherein like numerals denote like elements.

(2) FIG. 1 illustrates a perspective view of overhead stowage bins located in the interior of an aircraft, in accordance with various embodiments;

(3) FIGS. 2A and 2B illustrate a perspective view of an overhead stowage bin in a closed position and an open position, respectively, in accordance with various embodiments;

(4) FIG. 3 illustrates a schematic of an overhead stowage bin having a locking system configured to electrically control a latch; in accordance with various embodiments;

(5) FIG. 4 illustrates a perspective view of an overhead stowage bin having an electrically controlled door; in accordance with various embodiments; and

(6) FIG. 5 illustrates a schematic of an overhead stowage bin having an electrically controlled door; in accordance with various embodiments.

DETAILED DESCRIPTION

(7) The detailed description of exemplary embodiments herein makes reference to the accompanying drawings, which show exemplary embodiments by way of illustration. While these exemplary embodiments are described in sufficient detail to enable those skilled in the art to practice the exemplary embodiments of the disclosure, it should be understood that other embodiments may be realized and that logical changes and adaptations in design and construction may be made in accordance with this disclosure and the teachings herein. Thus, the detailed description herein is presented for purposes of illustration only and not limitation. The steps recited in any of the method or process descriptions may be executed in any order and are not necessarily limited to the order presented.

(8) Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact. Surface shading lines may be used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

(9) Disclosed herein is an overhead stowage bin having an electrically controlled locking system. In accordance with various embodiments, an electromechanical actuator (EMA) may be operationally coupled to a latch of the stowage bin. The EMA may be configured to actuate, or rotate, the latch in response to a signal from a bin controller. The bin controller may output commands to the EMA in response to receiving a signal from a passage input (e.g. a button, sensor, switch, touch pad, etc.). In various embodiments, the bin controller may be in communication with a master controller. The master controller may be configured to allow flight crew (e.g., pilot, co-pilot, flight attendants, etc.) to lock and unlock the overhead stowage bins. In various embodiments, one or more sensors located on or in the overhead stowage bin. The sensor may output information to the bin controller. In various embodiments, a display may be located on the overhead stowage bin door. The bin controller may control messages and/or information shown on the display.

(10) System program instructions and/or controller instructions may be loaded onto a tangible, non-transitory, computer-readable medium (also referred to herein as a tangible, non-transitory memory) having instructions stored thereon that, in response to execution by the controller, cause the controller to perform various operations. The term “non-transitory” is to be understood to remove only propagating transitory signals per se from the claim scope and does not relinquish rights to all standard computer-readable media that are not only propagating transitory signals per se. Stated another way, the meaning of the term “non-transitory computer-readable medium” and “non-transitory computer-readable storage medium” should be construed to exclude only those types of transitory computer-readable media which were found in *In Re Nuijten* to fall outside the scope of patentable subject matter under 35 U.S.C. § 101.

(11) Referring now to FIG. 1, an aircraft interior **10** including overhead stowage bins **100** located over seats **12** is shown, in accordance with various embodiments. Overhead stowage bins **100** may each include a door **102** configured to pivot about a hinge joint **104**. In various embodiments, a display **106** may be mounted to door **102**.

(12) With additional reference to FIGS. 2A and 2B, an overhead stowage bin **100** is illustrated in the closed position and the open position, respectively. Overhead stowage bin **100** may define a volume **108** configured to receive and store carry-on luggage. In accordance with various embodiments, a passenger input **110** may be located on door **102**. Passenger input **110** is in operable communication with a bin controller **112**. Passenger input **110** may be a push button,

touch pad, contactless sensor, radio frequency sensor, infrared sensor, ultra sound sensor, capacitive sensor, or any other device configured to output a signal to bin controller **112**. Passenger input **110** may communicate with bin controller **112** via wired, wireless, or mechanical connection. As described in further detail below, bin controller **112** is configured to output an electrical signal in response to receiving an “open bin” signal from passenger input **110**.

(13) Bin controller **112** may be electrically coupled to and in operable communication with an EMA **114**. EMA **114** is operably coupled to a latch **116** of overhead stowage bin **100**. In this regard, EMA **114** is configured to actuate (e.g., rotate, pivot, translate, etc.) latch **116** in response to receiving an electrical signal from bin controller **112**. Latch **116** is configured to secure door **102** in the closed position. In various embodiments, latch **116** may be configured to engage an anchor **118** affixed proximate a floor **120** of overhead stowage bin **100**. Anchor **118** may be located generally opposite hinge joint **104**. In accordance with various embodiments, EMA **114** may rotate latch **116** relative to anchor **118**. While FIG. 2B illustrates latch **116** attached to door **102** and anchor attached to floor **120**, it is contemplated and understood that latch **116** and anchor may be located in any location along door **102** and overhead stowage bin **100**. For example, in various embodiments, anchor **118** may be located along a sidewall **122** of overhead stowage bin **100**. Further, in various embodiments, anchor **118** may be attached to door **102** and latch **116** may be rotatably coupled to floor **120**. In various embodiments, overhead stowage bin **100** may include a mechanical override **124** configured to actuate latch **116**. Mechanical override **124** may be located generally out of view from passengers, which tends to prevent, or reduce a likelihood that, persons other than flight attendants will open overhead stowage bin **100** via mechanical override **124**.

(14) In various embodiments, overhead stowage bin **100** may include one or more sensors **130**. Sensors **130** may be employed individually, in groups, in series, and/or in parallel to obtain information related to overhead stowage bin **100**. For example, one or more sensors **130** may be configured to detect the presence of an object in the path of rotation of door **102**. One or more sensors **130** may be configured to detect a position of door **102** and/or a position of latch **116**. One or more sensors **130** may be configured determine a space availability with volume **108**. Stated differently, one or more sensors **130** may be configured to detect occupied area within volume **108**. One or more sensors **130** may be weight sensors, fire and/or smoke sensors, cabin pressure sensors, or any other desired sensor. Sensors **130** may be located along floor **120**, sidewalls **122**, door **102**, a ceiling **126** of overhead stowage bin **100**, and/or in any other desired location.

(15) With additional reference to FIG. 3, a schematic of a locking system **150** for overhead stowage bins **100** is illustrated. In accordance with various embodiments, locking system **150** may include the passenger input **110**, the bin controller **112**, the EMA **114**, the latch **116**, the mechanical override **124**, the sensors **130**, and the display **106** of each overhead stowage bin **100** within aircraft interior **10**. Bin controllers **112** may each include and communicate with one or more processors and one or more tangible, non-transitory storage medium(s) **152** and is capable of implementing locking system logic. The processor can be a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device (PLD), discrete gate or transistor logic, discrete hardware components, or a combination thereof. System program instructions and/or processor instructions may be loaded onto tangible, non-transitory computer-readable storage medium **152**. The system program instructions and/or processor instructions may, in response to execution by bin controller **112**, cause bin controller **112** to perform various operations. In particular, and as described in further detail below, the instructions may allow bin controller **112** to make operating decisions relative to EMA **114** and latch **116**. For example, bin controller **112** may be configured to determine whether to send an electrical signal to EMA **114** based on whether overhead stowage bin **100** is in a locked state or an unlocked state.

(16) In accordance with various embodiments, locking system **150** may further include a master controller **154**. Master controller **154** may be in operable communication with the bin controller

112 of each overhead stowage bin **100**. Master controller **154** may include and communicate with one or more processors and one or more tangible, non-transitory storage medium(s) **156** and is capable of implementing locking system logic. The processor can be a general purpose processor, a DSP, an ASIC, a FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or a combination thereof. System program instructions and/or processor instructions may be loaded onto tangible, non-transitory computer-readable storage medium **156**. The system program instructions and/or processor instructions may, in response to execution by master controller **154**, cause master controller **154** to perform various operations. In particular, and as described in further detail below, the instructions may cause master controller **154** to send lock commands to bin controllers **112** based on pilot or crew input received by master controller **154**.

(17) Locking system **150** is configured to allow a crewmember to lock overhead stowage bins **100**. For example, a pilot and/or flight attendant may desire to lock overhead stowage bins **100** during take-off or in the event of an emergency evacuation. In accordance with various embodiments, bin controller **112** may be configured to determine whether to send an electrical signal to EMA **114** based on commands received from master controller **154**. In accordance with various embodiments, master controller **154** may receive lock commands **164** and unlock commands **165** from the cockpit, for example, via a pilot input **180**. Pilot input **180** may be a push button, switch, touch pad, lever, contactless sensor, or any other device located in the cockpit and configured to allow a pilot to convey lock commands **164** and unlock commands **165** to master controller **154**. In response to receiving a lock command **164**, master controller **154** may output a lock command **162** to bin controller **112**. In response to receiving lock command **162**, bin controller **112** may convert to a locked state. In the locked state, bin controller **112** does not output electrical signals **160** to EMA **114**. The lack of electrical signals **160** prevents EMA **114** from actuating latch **116**. In this regard, in the locked state, latch **116** remains engaged to anchor **118**, with momentary reference to FIG. 2B, thereby preventing door **102** from opening.

(18) In accordance with various embodiments, master controller **154** may also receive lock commands **166** and unlock commands **167** from the cabin (e.g., from flight attendants) via a crew input **182**. Crew input **182** may be a push button, switch, touch pad, lever, contactless sensor, or any other device configured to allow a flight attendant to convey lock commands **166** and unlock commands **167** to master controller **154**. In response to receiving lock command **166**, master controller **154** may output lock command **162** to bin controller **112**. In response to receiving lock command **162**, bin controller **112** may convert to the locked state. In the locked state, bin controller **112** does not output electrical signals **160** to EMA **114**. The lack of electrical signals **160** prevents EMA **114** from actuating latch **116**. In this regard, latch **116** remains engaged to anchor **118**, with momentary reference to FIG. 2B, thereby preventing door **102** from opening.

(19) Locking system **150** may also be translated to the unlocked state via the pilot input **180** and/or the crew input **182**. For example, in accordance with various embodiments, master controller **154** may receive unlock command **165** from pilot input **180**. In response to receiving unlock command **165**, master controller **154** may output an unlock command **163** to bin controller **112**. In response to receiving unlock command **163**, bin controller **112** may convert to an unlocked state. In the unlocked state, bin controller **112** outputs electrical signals **160** to EMA **114** in response to receiving a signal (e.g., an open bin signal **170**) from passenger input **110**. In response to receiving electrical signals **160** from bin controller **112**, EMA **114** actuates latch **116**, thereby allowing door **102** to open.

(20) In accordance with various embodiments, master controller **154** may also receive unlock commands **167** from crew input **182**. In response to receiving an unlock command **167**, master controller **154** may output unlock command **163** to bin controller **112**. In response to receiving unlock command **163**, bin controller **112** may convert to the unlocked state. In the unlocked state, bin controller **112** outputs electrical signals **160** to EMA **114**, in response to bin controller **112**

receiving open bin signals **170** from passenger input **110**.

(21) In accordance with various embodiments, when overhead stowage bin **100** is in the closed position, a passenger or flight attendant may open door **102** and access volume **108** by pressing, touching, or otherwise causing passenger input **110** to output an open bin signal **170** to bin controller **112**. In response to receiving open bin signal **170**, bin controller **112** determines whether it is in the locked state or the unlocked state. If bin controller **112** determines it is in the unlocked, bin controller **112** outputs electrical signal **160** to EMA **114**, thereby causing EMA **114** to actuate latch **116**. If bin controller **112** determines it is in the locked, bin controller **112** does not output electrical signal **160** (i.e., bin controller **112** takes no action in response to receiving the open bin signal **170** from passenger input **110**).

(22) In accordance with various embodiments, bin controller **112** receives output **172** from sensors **130**. Bin controller **112** may output signals **173** correlating to the information received from sensors **130** to master controller **154**. Stated differently, sensors **130** may interface with bin controller **112** and bin controller **112** may communicate the information from sensors **130** to master controller **154**. Master controller **154** may process the information from sensors **130** and provide information to flight crew about the status of overhead stowage bins **100**. In various embodiments, bin controller **112** and/or master controller **154** may manipulate the information displayed on display **106** in response to information provided by sensors **130**. The information from sensors **130** may be employed to notify flight crew of an issue which requires attention, or to provide direct feedback through the display **106** to passengers using the overhead stowage bin **100**. The communication between sensors **130** bin controller **112** and between bin controller **112** and master controller **154** may be wireless based (such as Wi-Fi, Bluetooth, Near Field Communication, etc.) or wired based (such as Controller Area Network, Ethernet, etc.)

(23) Bin controller **112** may also be in communication with display **106**. Stated differently, bin controller **112** is operationally coupled to display **106**. Bin controller **112** is configured to send display commands **174** to display **106**. Display commands **174** may be based on output **172** from sensors and/or on commands (e.g., lock commands **162**, unlock commands **163**, or any other command) received from master controller **154**. In response to receiving display command **174**, display **106** generates and/or manipulates an image or message on display **106** to convey information corresponding to the display command **174**. Display **106** may be configured to output information via text, symbols, pictures, illumination of LEDs, or any other desired format. Display commands **174** may correspond to a locked status of latch **116** (e.g., unlocked state or locked state) or amount of available space within volume **108** of overhead stowage bin **100** (e.g., full, almost full, empty, 25% full, 50% full, 75% full, etc.). Conveying information related to the amount of available space within volume **108** may decrease passenger boarding times, as passengers and crew will be able to quickly and easily identify which overhead bins they can stow luggage in. In various embodiments, display **106** may output commercial advertisements, seat number information, directions to exits, time to destination, and/or any other desired message.

(24) Locking latch **116** and/or preventing release of latch **116** via electrical means provides a generally failsafe locking system. Locking system **150** can enhance the safety of overhead stowage bins by restricting access to baggage in the event of an aircraft evacuation or during periods of turbulence, when items can fall from the overhead stowage bins and injure passenger. Locking system **150** may decrease passenger boarding times by allowing crew to unlock the overhead bins located over the seat rows that are currently boarding and lock the overhead bins located over seat rows that have not yet boarded and/or lock the overhead bins that are full.

(25) With reference to FIG. 4, an overhead stowage bin **200** including an electric motor **202** is illustrated. Overhead stowage bin **200** may be similar to overhead stowage bin **100** in FIGS. 2A and 2B. Aircraft interior **10**, with momentary reference to FIG. 1, may include one or more overhead stowage bins **200** in place of one or more of the overhead stowage bins **100**. Electric motor **202** may be configured to actuate (e.g., rotate, pivot, etc.) a door **204** of overhead stowage

bin **200** about a hinge joint **206**. Overhead stowage bin **200** includes a display **208** and a passenger input **210**, similar to similar to display **106** and passenger input **110**, respectively, in FIG. 2A, FIG. 2B and FIG. 3. Passenger input **210** is in operable communication with a bin controller **212**. Passenger input **210** may communicate with bin controller **212** via wired, wireless, or mechanical connection. Bin controller **212** is operably coupled to electric motor **202**. Bin controller **212** is configured to control actuation of electric motor **202**, and thus the rotation of door **204** about hinge joint **206**, by outputting electrical signals to electric motor **202**. Bin controller **212** may configured to output an electrical signal configured to rotate door **204** in a first circumferential direction about hinge joint **206** (e.g., to an open position) in response to receiving a signal from passenger input **210**. For example, in response to receiving a signal from passenger input **210**, bin controller **212** may determine a position of door **204** and/or whether overhead stowage bin **200** is in a locked or unlocked state.

(26) With additional reference to FIG. 5, a schematic of a locking system **250** configured to control the opening and closing of door **204** is illustrated. In accordance with various embodiments, locking system **250** may include the passenger input **210**, the bin controller **212**, the electric motor **202**, sensors **230**, and the display **208** of each overhead stowage bin **200** within an aircraft interior. Bin controllers **212** may each include and communicate with one or more processors and one or more tangible, non-transitory storage medium(s) **252** and are capable of implementing locking system logic. System program instructions and/or processor instructions may be loaded onto tangible, non-transitory computer-readable storage medium **252**. The system program instructions and/or processor instructions may, in response to execution by bin controller **212**, cause bin controller **212** to perform various operations. In particular, and as described in further detail below, the instructions may allow bin controller **212** to make operating decisions relative to electric motor **202**. For example, bin controller **212** may be configured to determine whether to send an electrical signal to electric motor **202** based on whether the overhead stowage bin **200** is in a locked state or an unlocked state.

(27) In accordance with various embodiments, locking system **250** further includes a master controller **254**. Master controller **254** may be in operable communication with the bin controller **212** of each overhead stowage bin **200**. Master controller **254** may include and communicate with one or more processors and one or more tangible, non-transitory storage medium(s) **256** and is capable of implementing locking system logic. The system program instructions and/or processor instructions may, in response to execution by master controller **254**, cause master controller **254** to perform various operations. In particular, and as described in further detail below, the instructions may cause master controller **254** to send lock and unlock commands to bin controllers **212** based on pilot or flight attendant input received by master controller **254**.

(28) In accordance with various embodiments, bin controller **212** receives output **272** from sensors **230**. Bin controller **212** may output signals **273** correlating to the information received from sensors **230** to master controller **254**. Master controller **254** may process the information from sensors **230** and send commands to bin controller **212** and/or provide information to flight crew about the status of overhead stowage bins **200** based on the sensor output **272**. In various embodiments, bin controller **212** and/or master controller **254** may manipulate the information displayed on display **208** in response to information provided by sensors **230**. The information from sensors **230** may be employed to notify flight crew of an issue which requires attention, or to provide direct feedback through the display **208** to passengers using the overhead stowage bin **200**. Bin controller **212** may also be in communication with display **208**. Bin controller **212** is configured to send display commands **274** to display **208**, similar to bin controller **212** and display **106** in FIG. 3. The communication between sensors **230** bin controller **212** and between bin controller **212** and master controller **254** and between bin controller **212** and display **208** may be wireless based (such as Wi-Fi, Bluetooth, Near Field Communication, etc.) or wired based (such as Controller Area Network, Ethernet, etc.)

(29) Locking system **250** is configured to allow a crewmember to lock overhead stowage bins **200**. For example, a pilot and/or flight attendant may desire to lock overhead stowage bins **200** during take-off or in the event of an emergency evacuation. In accordance with various embodiments, master controller **254** may receive lock commands **264** and unlock commands **265** from the cockpit via a pilot input **280**, similar to pilot input **180** in FIG. 3. Master controller **254** may also receive lock commands **266** and unlock commands **267** from the cabin (e.g., from flight attendants) via a crew input **282**, similar to crew input **182** in FIG. 3. In response to receiving a lock command **264** and/or lock command **266**, master controller **254** may output a lock command **262** to bin controllers **212**. In response to receiving lock command **262**, bin controller **212** may convert to a locked state. In the locked state, bin controller **212** does not output electrical signals to electric motor **202**. The lack of electrical signals prevents electric motor **202** from actuating door **204**. In this regard, in the locked state, door **204** of overhead stowage bin **200** may be prevented from opening.

(30) Locking system **150** may also be translated to the unlocked state via pilot input **280** and/or the crew input **282**. For example, in accordance with various embodiments, master controller **254** may receive unlock command **265** from pilot input **180**. In response to receiving unlock command **265** and/or unlock command **267**, master controller **254** may output an unlock command **263** to bin controllers **212**. In response to receiving the unlock command **263**, bin controllers **212** may convert to an unlocked state. In the unlocked state, bin controller **212** outputs electrical signals **260a** or electrical signals **260b** (depending on the desired direction of rotation of door **204**) to electric motor **202**, in response to receiving an input signal (e.g., an open bin signal **270a** or a close bin signal **270b**) from passenger input **210**. In various embodiments, open commands and/or close commands configured to cause bin controllers **212** to output electrical signals configured to cause electric motor **202** to open door **204** and/or close door **204** may also be sent from crew input **282** and/or pilot input **280** via master controller **254**.

(31) In accordance with various embodiments, when overhead stowage bin **200** is in the closed position, a passenger or flight attendant may open door **204** by pressing, touching, or otherwise causing passenger input **210** to output an open bin signal **270a** to bin controller **212**. In response to receiving an open bin signal **270a**, bin controller **212** determines whether it is in the locked state or the unlocked state. If bin controller **212** determines it is in the unlocked state, bin controller **212** outputs electrical signal **260a** to electric motor **202**, thereby causing electric motor **202** to rotate about hinge joint **206** in a first circumferential direction. If bin controller **212** determines it is in the locked state, bin controller **212** does not output electrical signal **260a** (i.e., bin controller **212** takes no action in response to receiving the open bin signal **270a** from passenger input **210**). Overhead stowage bin **200** may also include a mechanical override **224** configured to actuate door **204** about hinge joint **206** should electric motor **202** fail.

(32) In various embodiments, in response to receiving open bin signal **270a**, bin controller **212** may also determine a position of door **204** based on output **272** from sensors **230**. In this regard, bin controller **212** may output electrical signal **260a** to electric motor **202** if bin controller **212** determines that it is in the unlocked state and that door **204** is not in a fully open position. If bin controller **212** determines that it is in the locked state or that door **204** is in the fully open position (i.e., door **204** cannot be rotated any farther in the first circumferential direction), bin controller **212** does not output electrical signal **260a** (i.e., bin controller **212** takes no action in response to receiving the open bin signal **270a** from passenger input **210**).

(33) In accordance with various embodiments, when overhead stowage bin **200** is in an open position (e.g., partially open or fully open), a passenger or flight attendant may close door **204** by pressing, touching, or otherwise causing passenger input **210** to output a close bin signal **270b** to bin controller **212**. In response to receiving close bin signal **270b**, bin controller **212** determines whether it is in the locked state or the unlocked state. If bin controller **212** determines it is in the unlocked state, bin controller **212** outputs electrical signal **260b** to electric motor **202**, thereby

causing electric motor **202** to rotate about hinge joint **206** in a second circumferential direction, opposite the first circumferential direction. If bin controller **212** determines it is in the locked state, bin controller **212** does not output electrical signal **260b** (i.e., bin controller **212** takes no action in response to receiving the close bin signal **270b** from passenger input **210**). In various embodiments, overhead stowage bin **200** is configured such that door **204** may also be closed by a passenger or crewmember physically applying a force to door **204** that rotates door **204** about hinge joint **206**. (34) In various embodiments, in response to receiving a close bin signal **270b**, bin controller **212** may also determine whether to send electrical signal **260b** to electric motor **202** based on a position of door **204**. Bin controller **212** may determine the position of door **204** based on output **272** from sensors **230**. For example, bin controller **212** may output electrical signal **260b** to electric motor **202** if, in response to receiving close bin signal **270b**, bin controller **212** determines that it is in the unlocked state and that door **204** is not in a fully closed position. If bin controller **212** determines that it is in the locked state or that door **204** is in the fully closed position (i.e., door **204** cannot be rotated any farther in the second circumferential direction), bin controller **212** does not output electrical signal **260b** (i.e., bin controller **212** takes no action in response to receiving the close bin signal **270b** from passenger input **210**). In various embodiments, one or more sensors **230** may be configured to detect the presence of an object in the path of door **204**. In this regard, bin controller **212** may determine not to send electrical signal **260b** (or electrical signal **260a**) in response to receiving output **272** from sensor **230** that indicates an object is in the path of door **204**. In this regard, controlling door **204** via electric motor **202** may increase safety and/or reduce probability of damage to luggage and/or door **204**.

(35) Benefits and other advantages have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, and any elements that may cause any benefit or advantage to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C.

(36) Systems, methods and apparatus are provided herein. In the detailed description herein, references to “various embodiments”, “one embodiment”, “an embodiment”, “an example embodiment”, etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(37) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112 (f), unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises”, “comprising”, or any other variation thereof, are intended to cover a

non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

Claims

1. An overhead stowage bin, comprising: a door configured for rotation about a hinge joint; an electric motor configured to rotate the door about the hinge joint; a bin controller electrically coupled to the electric motor; a first sensor in operable communication with the bin controller, wherein the first sensor is a cabin pressure sensor, and wherein the first sensor is configured to be located along at least one of a floor, a sidewall, the door, or a ceiling of the overhead stowage bin; and a display in operable communication with the bin controller, wherein the bin controller is configured to send a display command to the display, wherein the display command is based on an output received from the first sensor and corresponds to a cabin pressure within the overhead stowage bin, and wherein, in response to receiving the display command, the display generates at least one of an image or message to convey the cabin pressure within the overhead stowage bin.
 2. The overhead stowage bin of claim 1, further comprising a tangible, non-transitory memory configured to communicate with the bin controller, the tangible, non-transitory memory having instructions stored thereon that, in response to execution by the bin controller, cause the bin controller to perform operations comprising: receiving, by the bin controller, a signal from a passenger input; determining, by the bin controller, if the overhead stowage bin is in a locked state or an unlocked state; and determining, by the bin controller, whether to output an electrical signal to the electric motor based on whether the overhead stowage bin is in the locked state or the unlocked state.
 3. The overhead stowage bin of claim 2, wherein the electric motor is configured to rotate the door about the hinge joint in response to receiving the electrical signal from the bin controller.
 4. The overhead stowage bin of claim 3, further comprising a second sensor in operable communication with the bin controller, wherein the second sensor is configured to detect an object in a path of the door.
 5. The overhead stowage bin of claim 4, wherein the instructions further comprise: receiving, by the bin controller, an output from the second sensor; and determining, by the bin controller, whether to forward the electrical signal to the electric motor based on the output from the second sensor.
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